

Exercise 1.

Use the fact that normal pdf is well define, calculate the integral

$$\int_{-\infty}^{\infty} e^{-\frac{(x+2)^2}{6}} dx = ?$$

Exercise 2.

A card is picked at random from a shuffled card deck for 500 consecutive times. (a) What is the approximate probability that a red card will be picked between 250 and 300 times inclusively? (b) What is the approximate probability that an even-numbered card will be picked more than 200 times?

Exercise 3.

A small airline makes five flights a day from Chicago to Denver. The number of passengers on each flight is approximated by an exponential random variable with mean 20. A flight makes money if it has more than 25 passengers. Find the probability that at least one flight a day makes money. Assume that the numbers of passengers on different flights are independent.

Exercise 4.

The new outpost on Mars extracts water from the surrounding soil. There are 13 extractors. Each extractor produces water with a random efficiency that is uniformly distributed on $[0,1]$. The outpost operates normally if fewer than three extractors produce water with efficiency less than 0.25. If the efficiencies are independent, find the probability that the outpost operates normally.

Exercise 5.

$X \sim N(3, 16)$, $Y \sim N(\mu_Y, 9)$ and $P(X + Y > 8.5) = 0.0764$. Find the value of μ_Y .